Project Proposal (Group 8)

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Enhancing Image Classification Performance Using GAN-Based Synthetic Data Augmentation

1. Introduction

Deep learning models have achieved remarkable success in computer vision tasks such as image classification. However, these models often require large amounts of labeled training data to achieve high performance and generalization. In many real-world scenarios, obtaining large, labeled datasets can be expensive, time-consuming, or limited by privacy concerns.

Generative Deep Learning provides a promising solution to this problem by enabling the creation of synthetic data that closely resembles real data. One of the most influential approaches in this area is the **Generative Adversarial Network (GAN)**. GANs consist of two neural networks, a generator and a discriminator, which compete with each other during training. The generator attempts to create realistic samples while the discriminator learns to distinguish between real and generated data.

This project investigates whether synthetic images generated by GANs can be used as **data augmentation** to improve the performance of image classification models.

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2. Problem Statement

Many machine learning models suffer from reduced performance when training data is limited. Traditional data augmentation methods such as rotation, flipping, and cropping help increase dataset diversity but may not introduce entirely new samples.

Generative models such as GANs can produce **new synthetic data samples** that follow the same distribution as the original dataset. However, it is still an open question whether such synthetic data can significantly improve classification performance.

The objective of this project is to study the effectiveness of GAN-generated synthetic images as a form of **data augmentation for image classification tasks**.

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3. Research Questions

This project will explore the following research questions:

1. Can GAN-generated synthetic images improve the performance of image classification models when combined with real training data?

2. How does classification performance compare when training with:
    - only real data,
    - real data combined with synthetic data, and
    - synthetic data only?
3. What is the quality of the generated images and how closely do they resemble real samples from the dataset?

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4. Related Work

Generative models have become a major research area within generative deep learning. The GAN framework proposed by Generative Adversarial Networks[1] introduced an adversarial training process in which a generator network learns to produce synthetic samples while a discriminator network attempts to distinguish between real and generated data.

Later work extended GANs to conditional settings. For example, Conditional Generative Adversarial Nets[2] proposed conditioning the generator on class labels, enabling the generation of images belonging to specific categories.

Recent research has explored the use of generative models for data augmentation. For example, Antoniou et al. proposed Data Augmentation GANs[3] that generate synthetic training samples to improve classifier generalization. Improvements in GAN architectures such as StyleGAN[4] have also significantly enhanced the realism and quality of generated images.

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5. Proposed Methodology

The project will involve the following steps.

5.1 Dataset Selection

We will use a widely studied benchmark dataset for image classification. Possible datasets include:

- MNIST handwritten digit dataset
- Fashion-MNIST dataset

These datasets are suitable because they are well understood and allow efficient training of generative models.

5.2 Training a Generative Model

A Generative Adversarial Network will be trained on the selected dataset. The GAN will consist of:

- **Generator network**: produces synthetic images from random noise
- **Discriminator network**: distinguishes between real images and generated images

Through adversarial training, the generator learns to produce increasingly realistic images.

5.3 Synthetic Data Generation

Once the GAN model is trained, synthetic images will be generated to create an additional dataset. These generated samples will then be used to augment the original training data.

5.4 Training Image Classifiers

A convolutional neural network (CNN) classifier will be trained under three different conditions:

1. **Baseline Model** - trained using only real training data.
2. **Augmented Model** - trained using real data plus GAN-generated synthetic data.
3. **Synthetic Data Model** - trained using only synthetic data.

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6. Evaluation Metrics

The classification models will be evaluated using standard metrics including:

- Accuracy
- Precision and Recall
- F1-score
- Confusion Matrix

In addition, the quality of the generated images will be visually inspected to assess their similarity to real samples.

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7. Tools and Implementation

The project will be implemented using the following tools:

- Python
- PyTorch
- Google Colab or GPU-enabled environment

Libraries to be used include:

- PyTorch
- torchvision
- NumPy

- Matplotlib

These tools will support model training, data generation, visualization, and evaluation.

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8. Expected Contributions

The expected outcomes of this project include:

1. A trained generative model capable of producing synthetic images.
2. A comparison of classification performance across different training scenarios.
3. An analysis of the effectiveness of generative data augmentation.
4. Insights into the benefits and limitations of using GANs for improving classification performance.

9. References

[1] Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). *Generative Adversarial Networks*. Advances in Neural Information Processing Systems (NeurIPS 27). Neural Information Processing Systems Foundation.

[2] Mirza, M., & Osindero, S. (2014). *Conditional Generative Adversarial Nets*. arXiv preprint arXiv:1411.1784. Published in the International Conference on Learning Representations (ICLR) Workshop.

[3] Antoniou, A., Storkey, A., & Edwards, H. (2017). *Data Augmentation Generative Adversarial Networks*. arXiv preprint arXiv:1711.04340.

[4] Karras, T., Laine, S., & Aila, T. (2020). *Analyzing and Improving the Image Quality of StyleGAN*. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).